Appendix A - Abrolhos Marine Park (Big Bank) Natural Values Report

South-west Marine Parks Network

Table of contents

Natural Values 1
- Sampling summary . . . 2
- Benthic ecosystem extent . . . 3
  - Observations . . . 3
  - Dominant habitat types . . . 4
  - Habitat distribution and probability . . . 5
  - Functional reef distribution and probability . . . 6
- Benchmarks of benthic natural values . . . 7
  - Sand . . . 7
  - Sessile invertebrates . . . 8
- Benchmarks of demersal fish natural values . . . 9
  - Survey effort . . . 9
  - Species richness . . . 10
  - Total abundance . . . 12
  - Large Reef Fish Index* . . . 14
  - Reef fish thermal index . . . 16

Natural Values

Understanding of natural values

Sampling summary

Table 1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone.

					Habitat	Fish	
Date	**Campaign name**	**Method**	**Areas**	**Samples**	**Count**	**Count**	**Length**
May 2021	2021-05 Abrolhos BOSS	BOSS	NPZ	29	29	-	-
May 2021	2021-05 Abrolhos stereo-BRUVs	BRUV	NPZ	26	26	26	24

Benthic ecosystem extent

Observations

Figure 1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. Australian Marine Park boundaries are shown. Bathymetric contours at 30, 70 and 200 m are shown.

Dominant habitat types

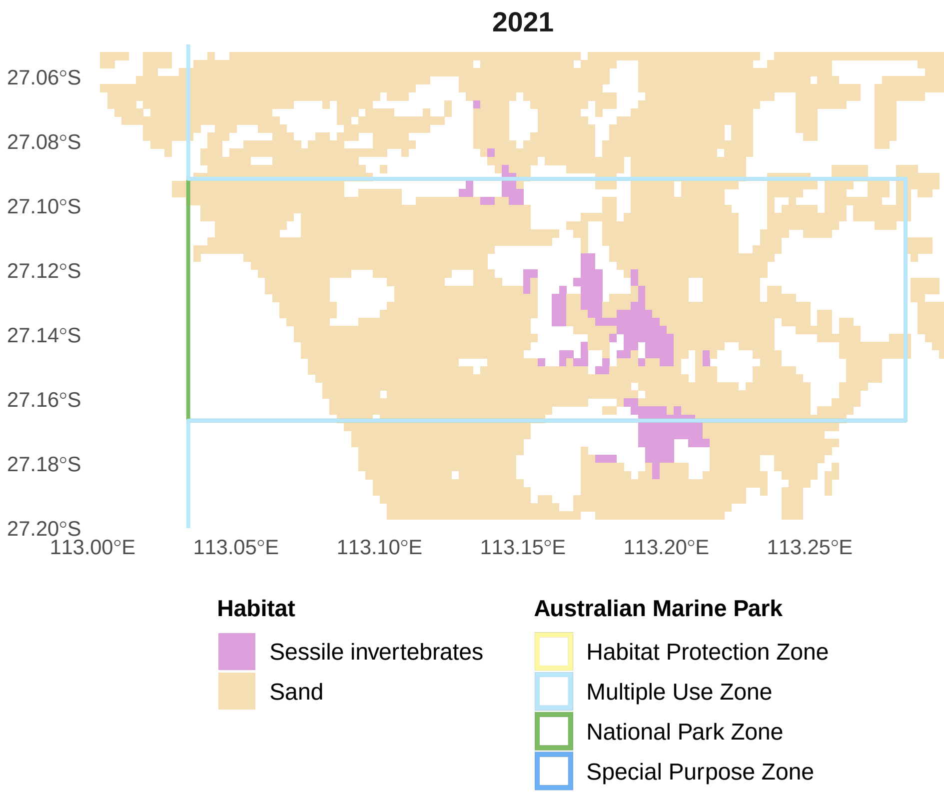

Figure 2: Predicted dominant habitat type from stereo-BRUV groundtruthing. Australian Marine Park boundaries are shown. Bathymetric contours at 30, 70 and 200 m are shown.

Habitat distribution and probability

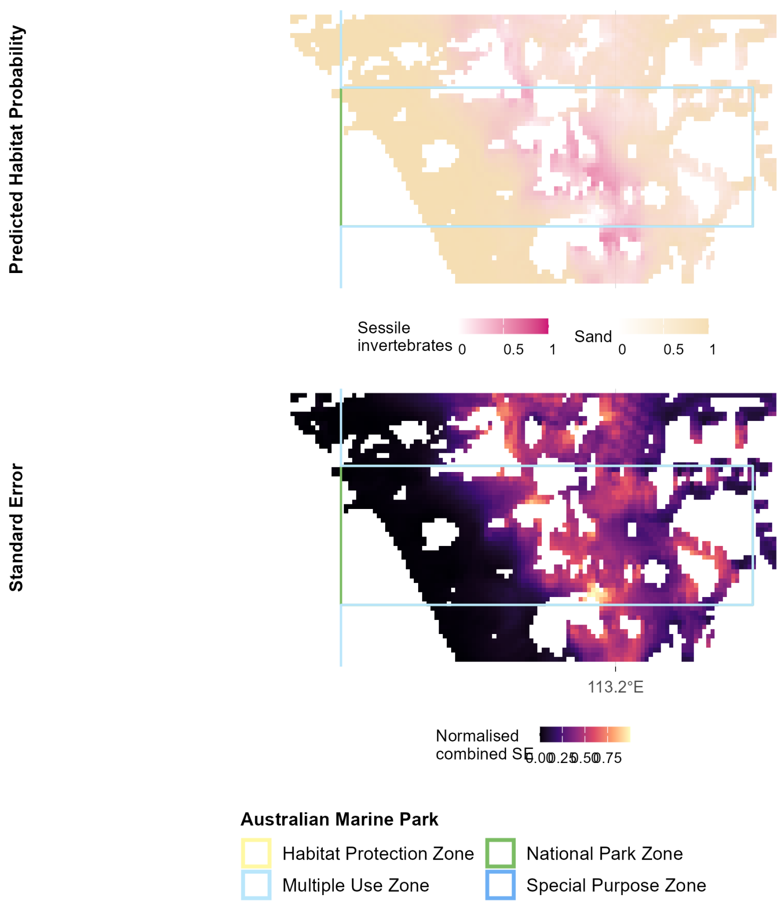

Figure 3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. Australian Marine Park boundaries are shown. Bathymetric contours at 30, 70 and 200 m are shown.

Functional reef distribution and probability

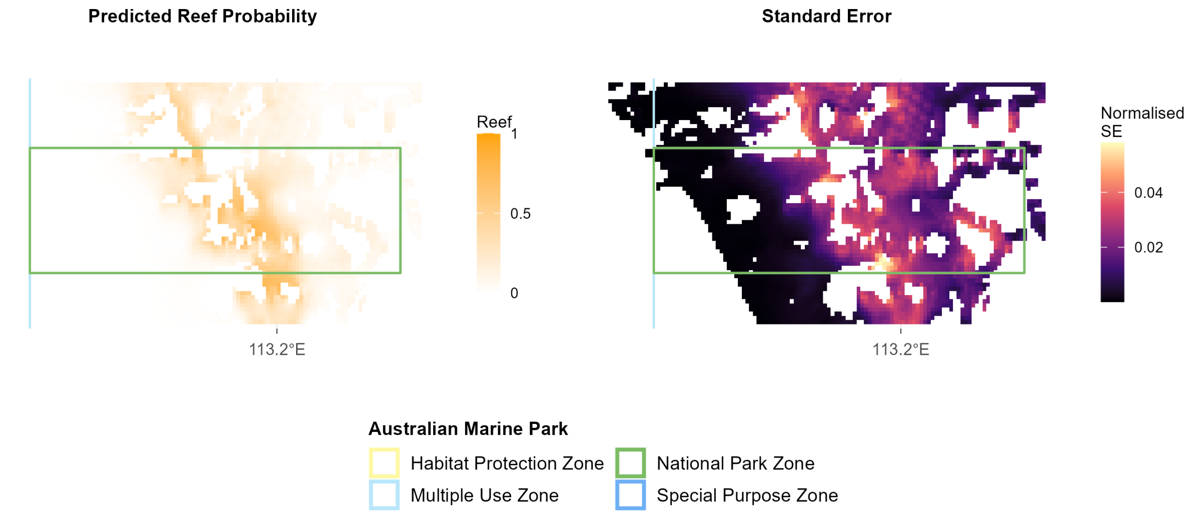

Figure 4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Australian Marine Park boundaries are shown. Bathymetric contours at 30, 70 and 200 m are shown.

Benchmarks of benthic natural values

Sand

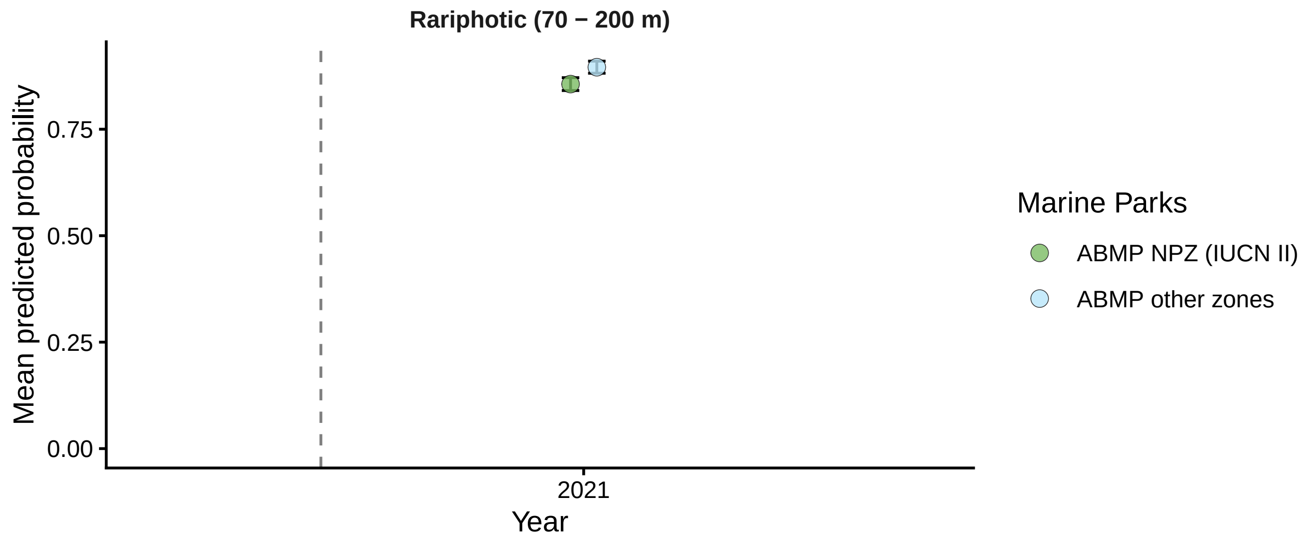

Figure 5.1: Sand: temporal plots by depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

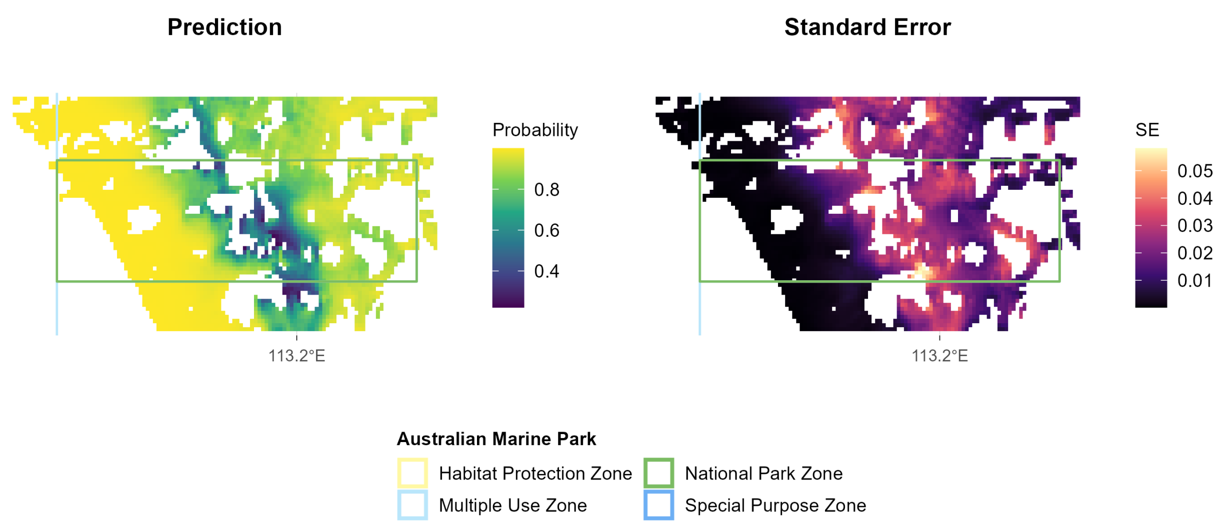

Figure 5.2: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). Australian Marine Park boundaries are shown. Bathymetric contours at 30, 70 and 200 m are shown.

Sessile invertebrates

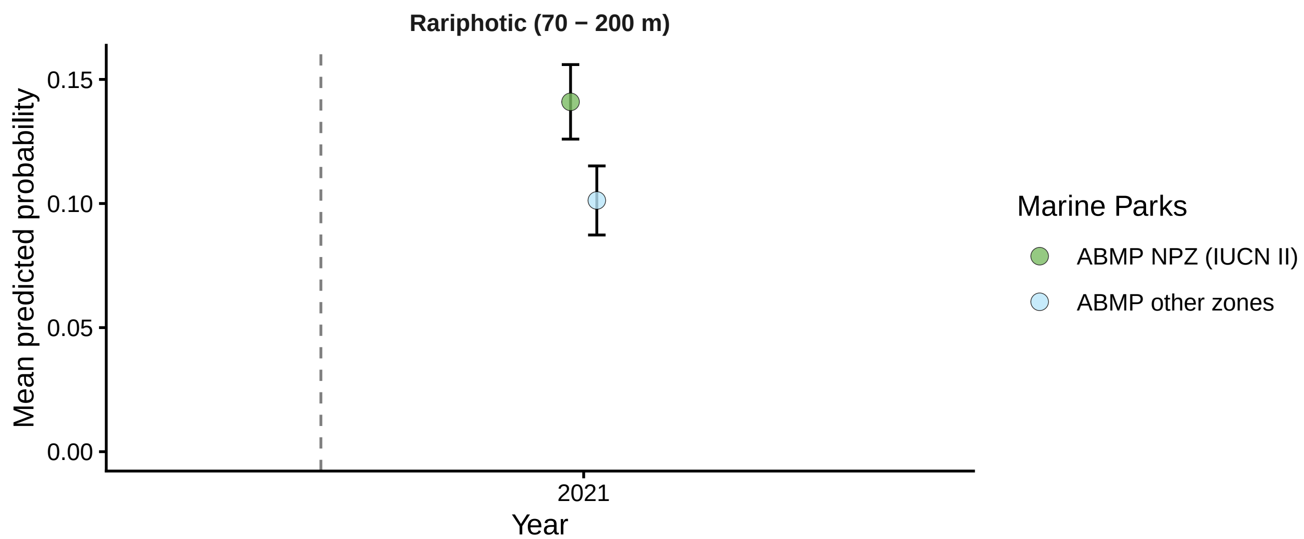

Figure 6.1: Sessile invertebrates: temporal plots by depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

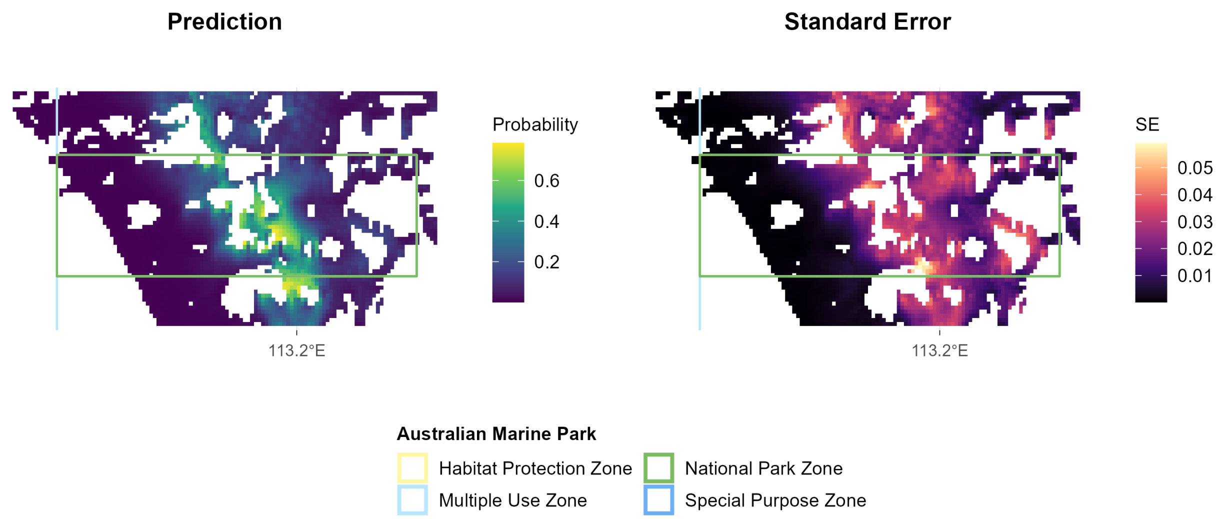

Figure 6.2: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). Australian Marine Park boundaries are shown. Bathymetric contours at 30, 70 and 200 m are shown.

Benchmarks of demersal fish natural values

Survey effort

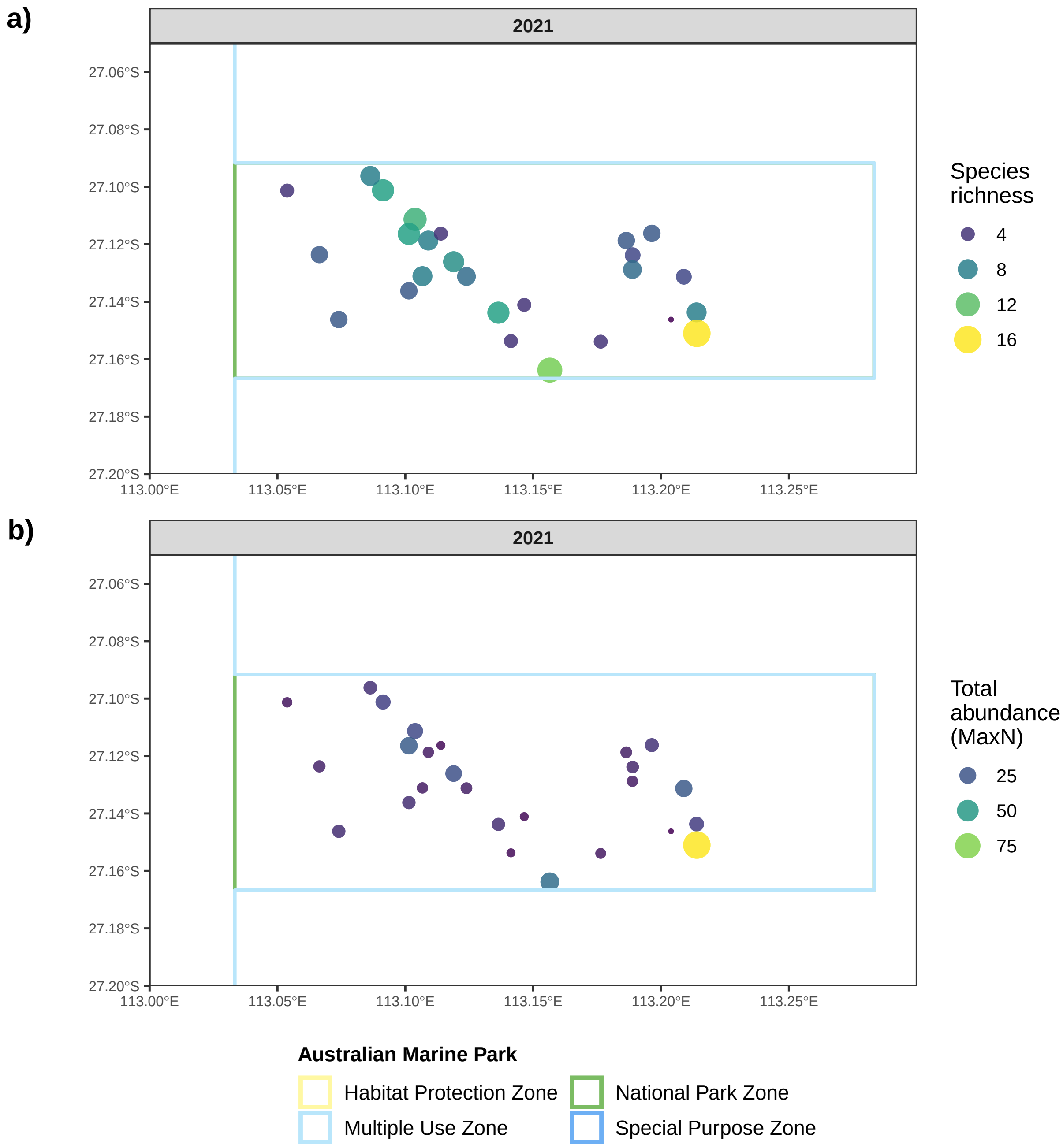

Figure 7: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Australian Marine Park boundaries are shown.

Species richness

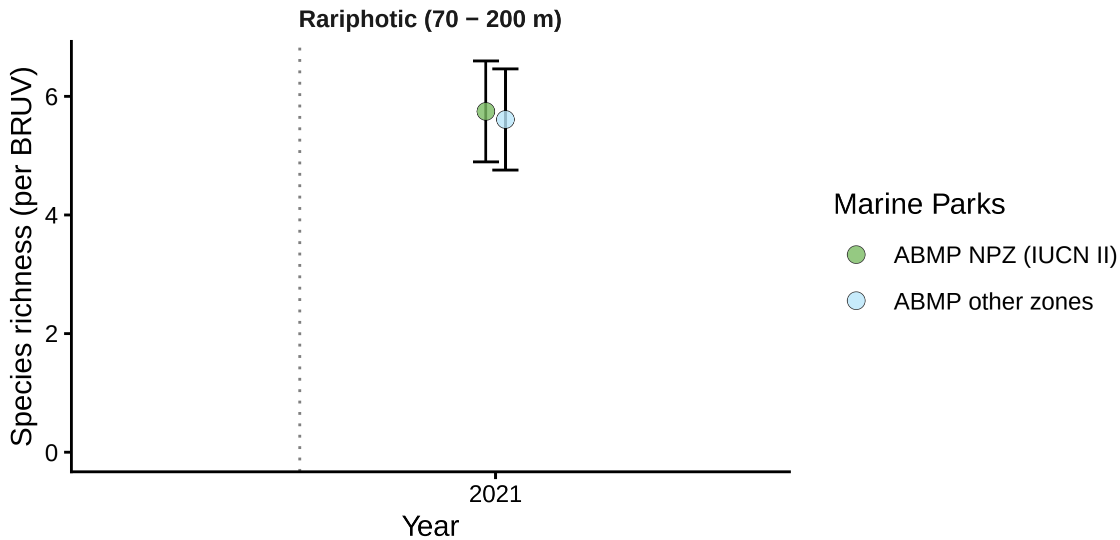

Figure 8.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

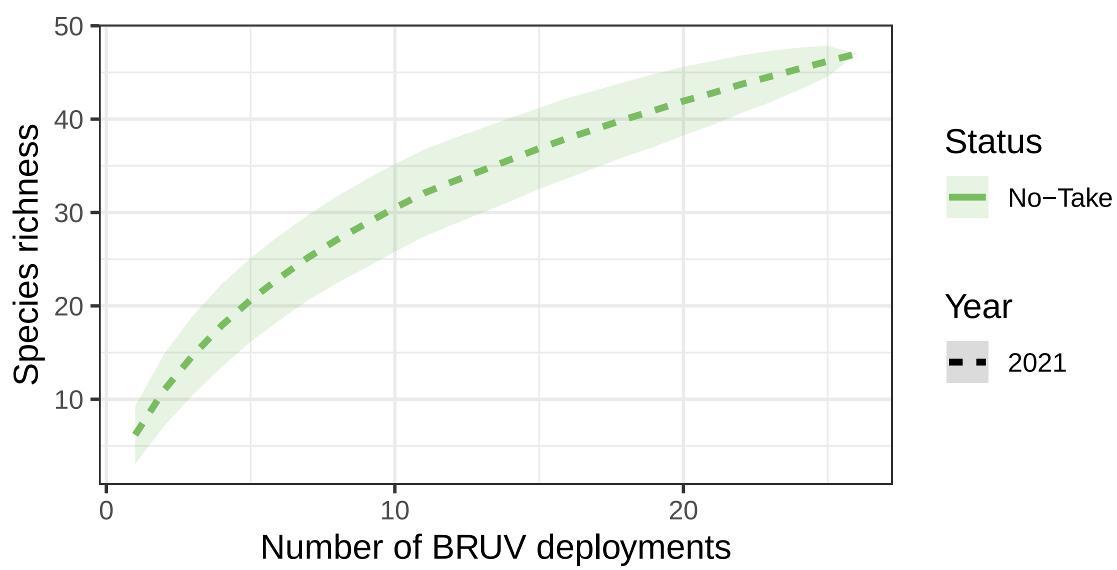

Figure 8.2: Demersal fish - Species richness: species accumulation curve from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

Species richness

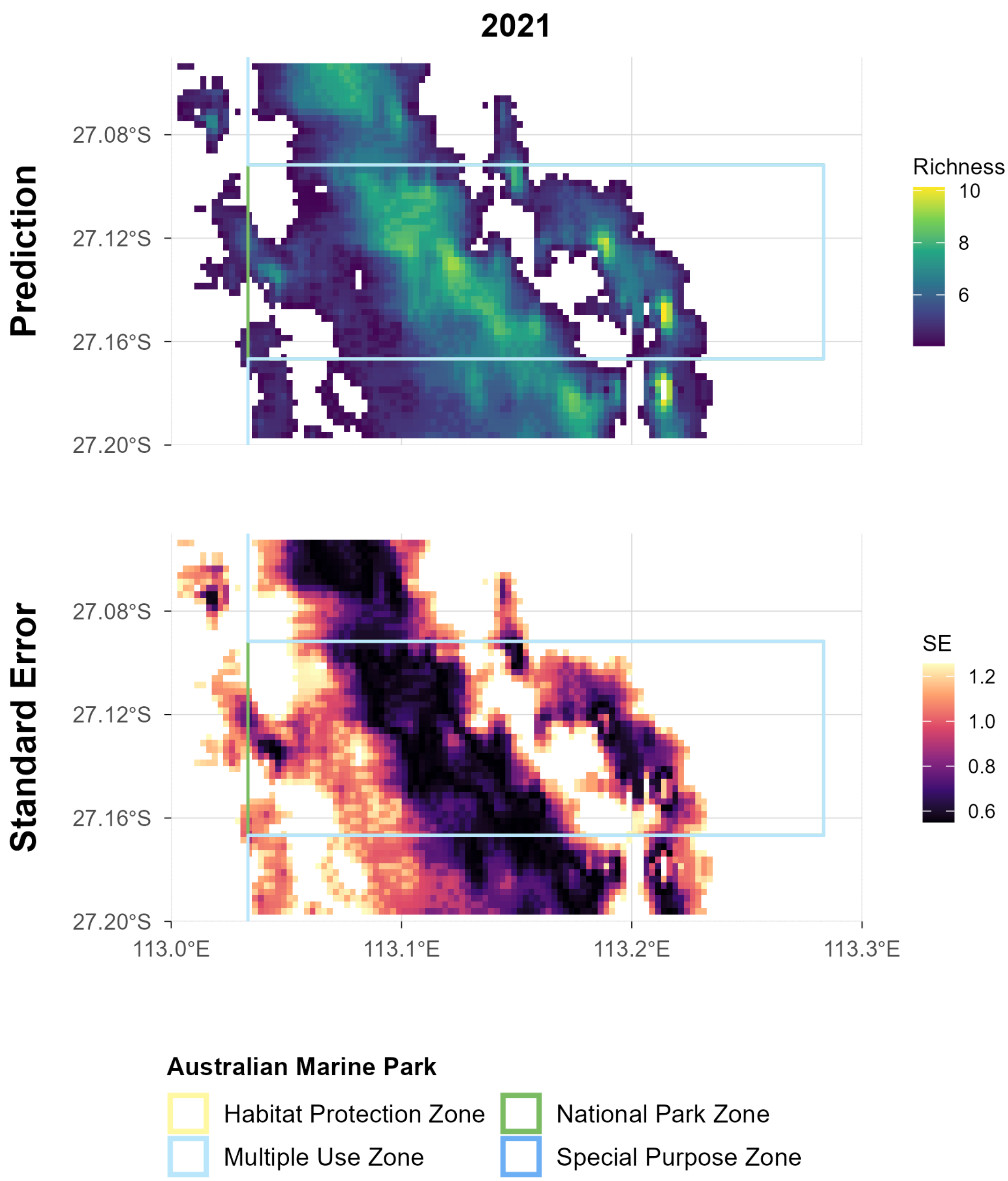

Figure 8.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Australian Marine Park boundaries are shown. Bathymetric contours at 30, 70 and 200 m are shown.

Total abundance

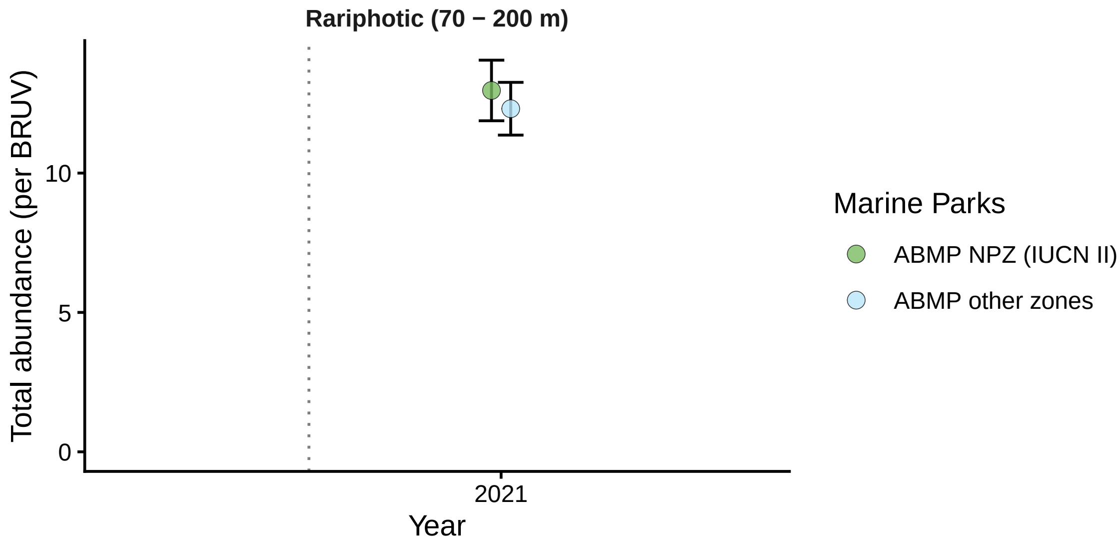

Figure 9.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

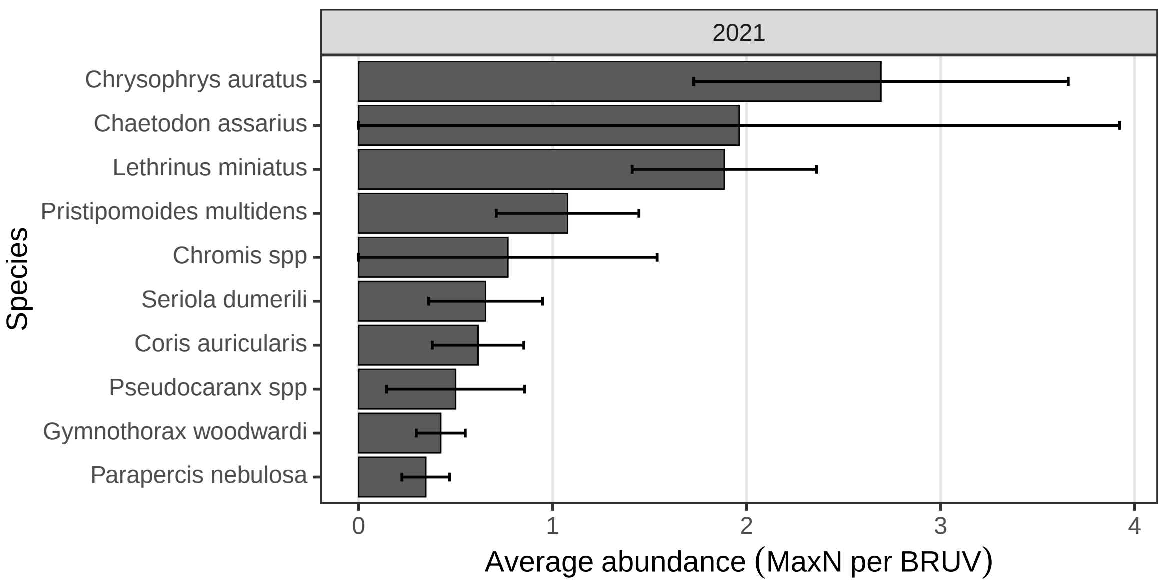

Figure 9.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

Total abundance

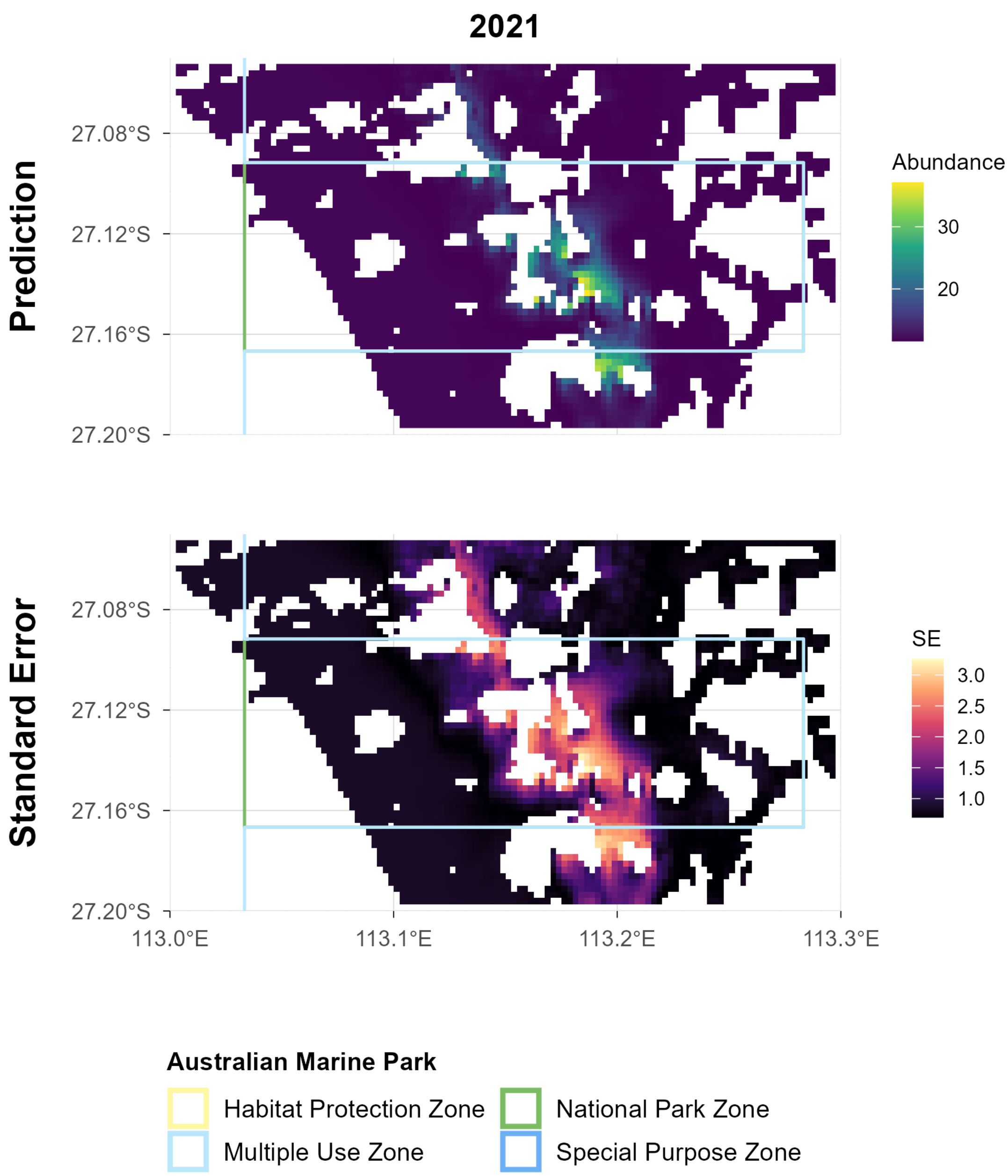

Figure 9.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Australian Marine Park boundaries are shown. Bathymetric contours at 30, 70 and 200 m are shown.

Large Reef Fish Index*

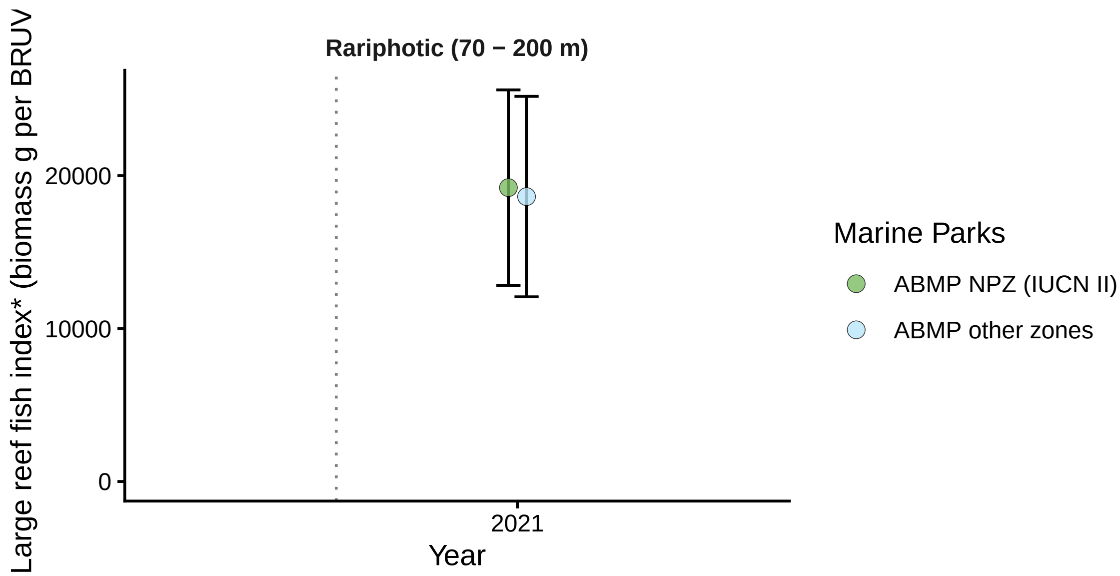

Figure 10.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth). *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

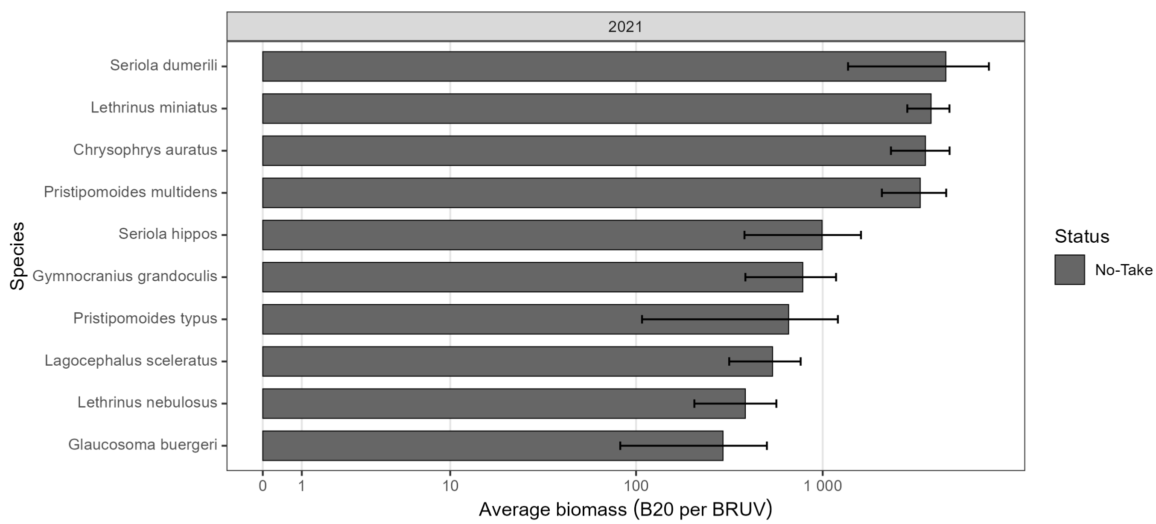

Figure 10.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

Large Reef Fish Index*

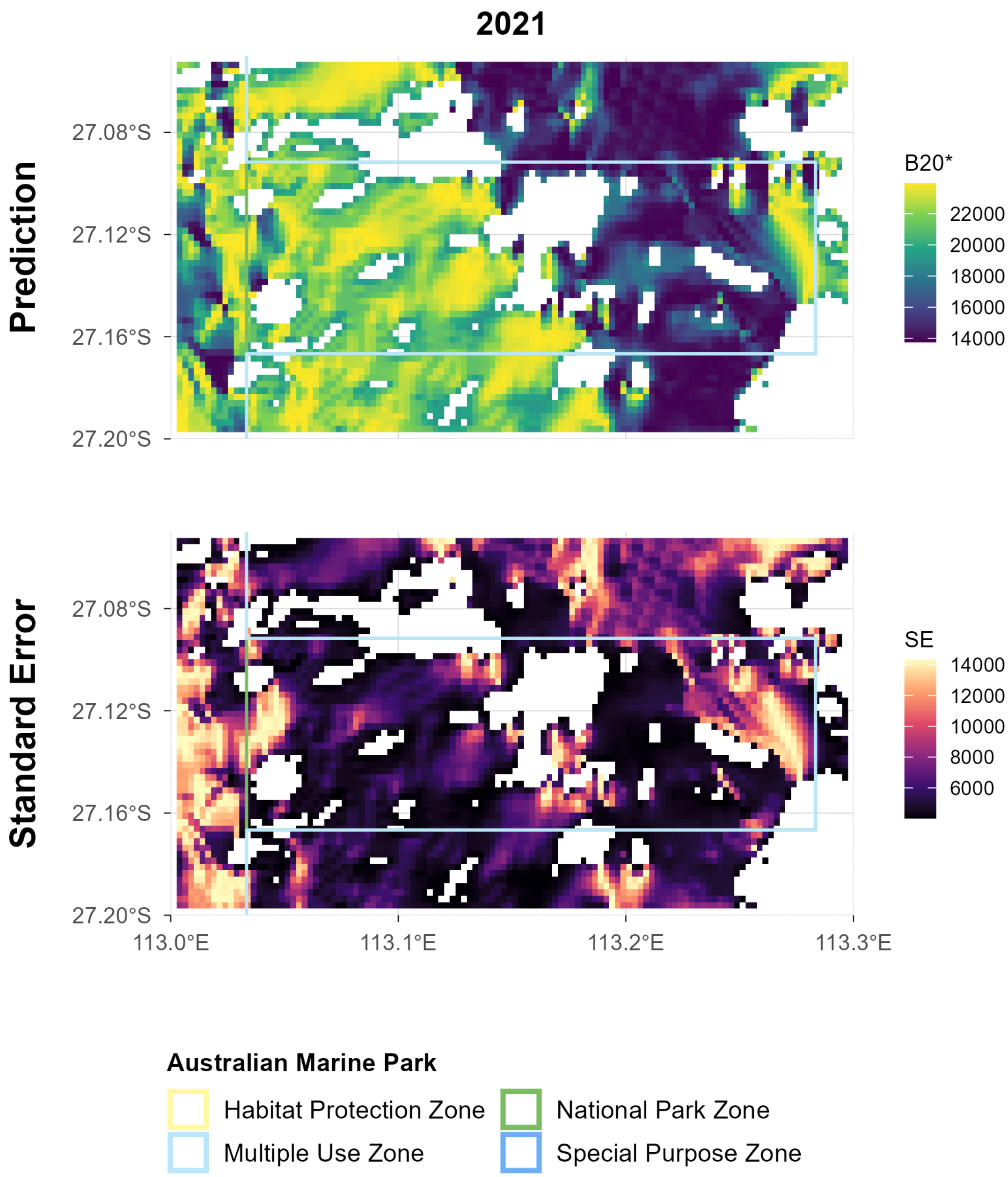

Figure 10.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Australian Marine Park boundaries are shown. Bathymetric contours at 30, 70 and 200 m are shown.

Reef fish thermal index

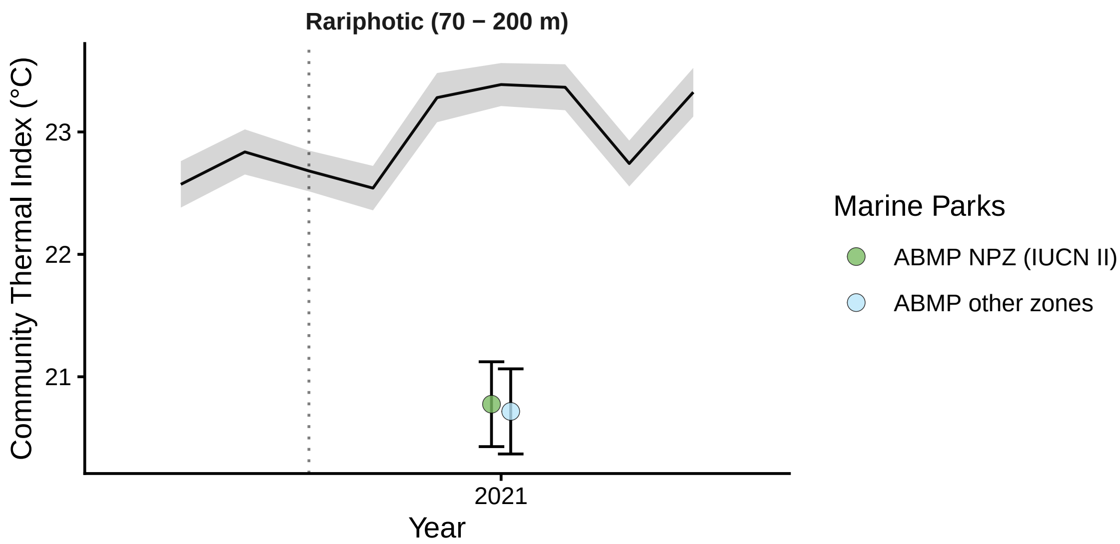

Figure 11.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

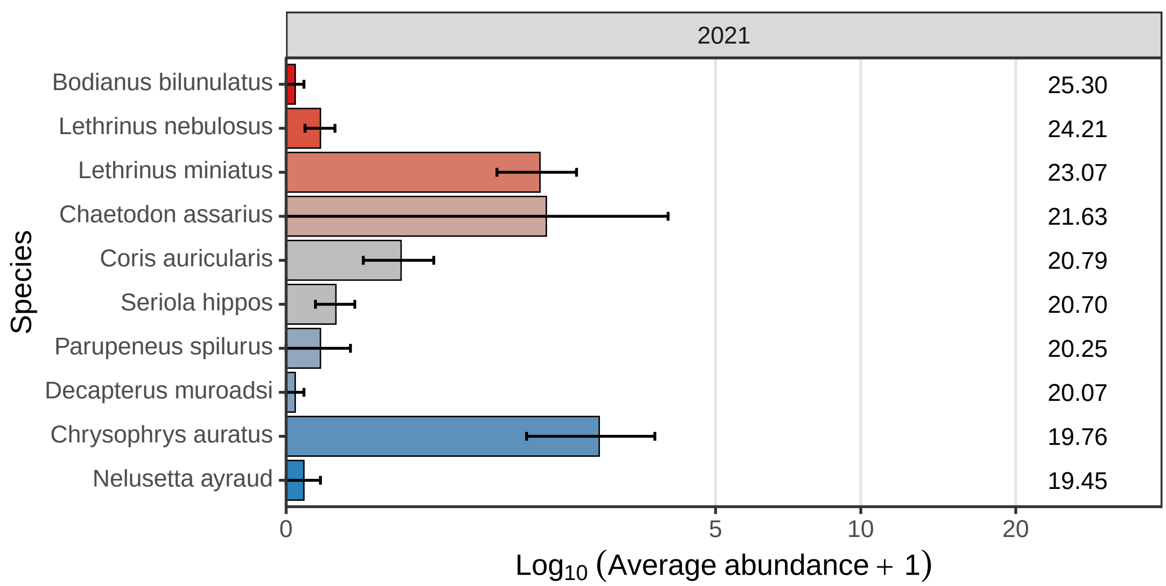

Figure 11.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. ABMP = Abrolhos Marine Park (Commonwealth).

Reef fish thermal index

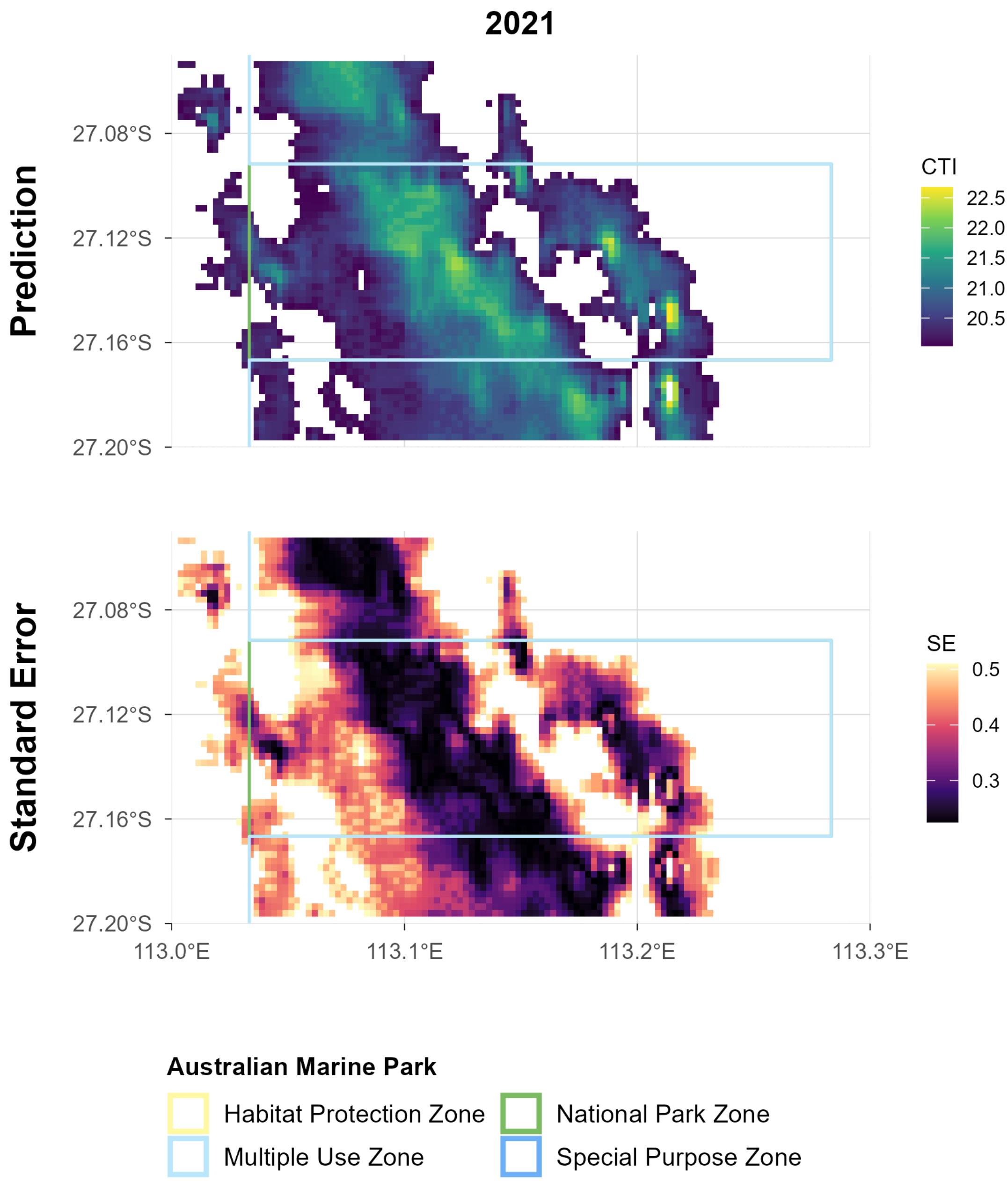

Figure 11.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Australian Marine Park boundaries are shown. Bathymetric contours at 30, 70 and 200 m are shown.